

March 3rd 2026

Alp Efe Kılıçarslan, 22402390

EEE 202-1

LAB-02: Second-Order RLC Circuit's Transient Response to Step Input and Sinusoids

Introduction:

The goal of this assignment is to analyze the transient and steady-state response of a second-order RLC circuit to both step and sinusoidal excitations. We will construct a circuit with two non-power supply related branches. One branch will be purely resistive, while the other will contain a series LC circuit with a series but small resistor, which is to model the resistive component of the inductor. Power will be supplied to the circuit in pulse-like steps and the capacitor voltage will be probed and recorded. In the preliminary step, analytical derivations will be made by modeling said circuit with a second order differential equation and solving it for rectangular pulses and n-sinusoidal pulses for different time intervals. The solutions to said systems will be plotted using MATLAB, and simulated using LTSpice. After this, the experiment will be conducted in real life to compare our findings with real life measurements.

Analysis:

The general analytical solution of the circuit (Figure A) will be derived using linear algebraic methods and by utilizing element parameters. The parameters of the resistor ($V = IR$), capacitor ($i_C = C \frac{dv_C}{dt}$) and the inductor ($V_L = L \frac{di_L}{dt}$) will be used. Before solving the second order circuit's solution, we will do some operations to make the analysis easier.

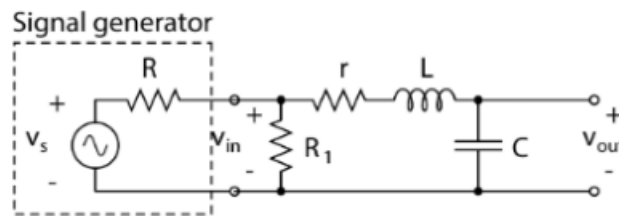


Figure A: Circuit that will be analyzed

First of all, we will denote the values for the main circuit in accordance to the Lab 2 RLC Assignment document. For this case, the capacitance was given as 12 nF, the inductance was 10 μ H, the resistance was 2 Ω and the signal generator's source impedance was 50 Ω . According to the Lab 2 document, the

inductors series resistance was to be calculated using the relation $r = \sqrt{\frac{f_r}{10\text{KHz}}} r_{nom}$. For our inductor, r_{nom} was given as 0.1Ω and the resonant frequency f_r was given as 459 KHz. Through these values, r was found to be approximately 0.6775Ω . For the time durations that will be picked in the upcoming preliminary work sections, the resonant frequency is 459 KHz and the resonant period is 2.1766 microseconds.

After these values were noted, the circuit was transformed using Thevenin equivalency for the ease of analysis. By putting the source voltage supply, the source resistor and the arbitrary resistor were combined to a single unit. Since the resistors of R and R_1 are in parallel and have a value of 50 and 2 Ohms respectively, the Thevenin resistance was found to be approximately 1.9231 Ohms; and the Thevenin voltage was found to be $0.0385 V_s$ through the voltage divider equation. By denoting said resistance and voltage as R_x and V_x respectively, we can redraw the circuit (Figure B) for easier analysis for V_{out} .

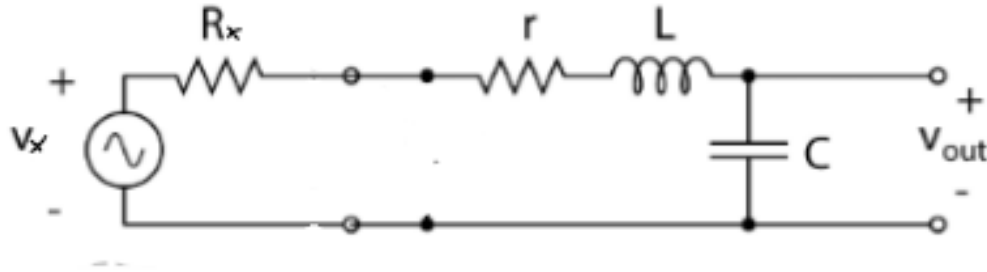


Figure B: Described Thevenin Equivalent Circuit

In our transformed circuit, we have a singular mesh, which means that the current for all of the components is the same. By taking the previously mentioned element parameters and Kirchoff's Voltage Law, we can make the following derivation:

$$V_{R_{net}}(t) + V_L(t) + V_C(t) = V_x(t) \quad (1)$$

$$(R_x + r)i_{net}(t) + L \frac{di_{net}(t)}{dt} + V_C(t) = V_x(t) \quad (2)$$

$$(R_x + r)C \frac{dV_C(t)}{dt} + LC \frac{d^2V_C(t)}{dt^2} + V_C(t) = V_x(t) \quad (3)$$

$$\frac{d^2V_C(t)}{dt^2} + \frac{(R_x + r)}{L} \frac{dV_C(t)}{dt} + \frac{V_C(t)}{LC} = \frac{V_x(t)}{LC} \quad (4)$$

Which is a second-order non-homogenous linear differential equation. This means that the general solution -which will be utilized for different input voltages- is the sum of the homogenous case's solution and a particular solution based on the forcing function, which is $\frac{V_x(t)}{LC}$ in this case. For future requirements,

the homogenous solution should be derived and used as it is not dependent on the forcing function. To do this, we can insert the component values that were assigned and calculated into equation (4), writing the characteristic equation, finding the roots and solving. These steps can be seen as:

$$\frac{d^2V_C(t)}{dt^2} + 2.6006 \times 10^5 \frac{dV_C(t)}{dt} + 8.3333 \times 10^{12} V_C(t) = 0 \quad (5)$$

$$\lambda^2 + 2.6006 \times 10^5 \lambda + 8.3333 \times 10^{12} = 0 \quad (6)$$

$$\lambda_{1,2} = -1.3003 \times 10^5 \pm 2.8838 \times 10^6 i \text{ (by the quadratic equation)} \quad (7)$$

$$V_C(t) = Ae^{\lambda_1} + Be^{\lambda_2} V \quad (8)$$

$$V_C(t) = e^{-1.3003 \times 10^5 t} [A \cos(2.8838 \times 10^6 t) + B \sin(2.8838 \times 10^6 t)] V \quad (9)$$

Where A and B are arbitrary constants depending on the initial conditions such as initial capacitor voltage and current, and for time in seconds. By deriving this equation, we have solidified a strong basis for the rest of the preliminary tasks.

Preliminary Part 1 (Quality Factor Calculation):

Using the Thevenin equivalent circuit in Figure B, we can use the formula for the quality factor Q, which is known as $Q = \frac{\omega_0 L}{R}$ for series RLC circuits, where ω_0 is the resonant angular frequency, L is the inductance and R is the net resistance. Plugging in our derived values ($\omega_0 = 2\pi f_r$, $L = 10 \mu\text{H}$, $R = 2.6006 \text{ Ohm}$), we can see that the quality factor is approximately **11.0897**. It should be noted that this value depends on f_r , L and all the resistance values since net resistance is dependent on r, R and R_1 .

Preliminary Part 2 (Step Pulse Response):

For this part, we need to solve equation (4). As is previously mentioned, the solution will be the combination of the homogenous case and a particular solution. We defined V_x to be $0.0385 V_s$, and for this specific case, V_s is a constant 8 Volts. This means that V_x is approximately 308 millivolts. For the particular solution $V_{Cp}(t)$, we can propose it to be of a constant solution since the derivative of a constant is zero. Plugging in $V_{Cp}(t) = P$ where P is a constant into equation (4), we get $\frac{P}{LC} = \frac{V_x(t)}{LC}$, which means that P is equal to 308 millivolts. Therefore, we have the general solution:

$$V_C(t) = e^{-1.3003 \times 10^5 t} [A \cos(2.8838 \times 10^6 t) + B \sin(2.8838 \times 10^6 t)] + 0.308 V \quad (10)$$

For the rectangular pulse case during the pulse and time in seconds. Now we need to find A and B from the initial conditions. By definition from the lab report, we have zero capacitor voltage and zero capacitor current for $t < 0$. By plugging in $t=0$ into equation (10), we can determine that A is equal to the particular solution, which is -0.308 Volts since initial capacitor voltage is zero. Taking the derivative of equation (10), we get:

$$\frac{dV_C(t)}{dt} = (e^{-1.3003 \times 10^5 t} \times 2.8838 \times 10^6) [-A \sin(2.8838 \times 10^6 t) + B \cos(2.8838 \times 10^6 t)] - (1.3003 \times 10^6 \times e^{-1.3003 \times 10^5 t}) [A \cos(2.8838 \times 10^6 t) + B \sin(2.8838 \times 10^6 t)] V \quad (11)$$

Which, by the element parameter of $i_C = C \frac{dV_C}{dt}$, is equal to $\frac{i_C}{C}$, which should be zero at $t=0$. By this fact, we can determine B to be approximately -0.0139 Volts. Finally, we can plug in these to find:

$$V_C(t) = -e^{-1.3003 \times 10^5 t} [0.308 \cos(2.8838 \times 10^6 t) + 0.0139 \sin(2.8838 \times 10^6 t)] V + 0.308 V \quad (12)$$

During the pulse duration for t in seconds. Since we are asked for a unit function voltage of 8V, we can use a long pulse for the duration of the simulation for LTSpice, and we can just use equation (12) in MATLAB.

After the analytical derivations, the circuit was formed in LTSpice to measure the output voltage for $t > 0$. After it was formed in accordance to the specifications, a transient analysis was performed and the capacitor voltage was plotted (Figure C). Additionally, the analytical solution was written in MATLAB and the plot was generated (Figure D).

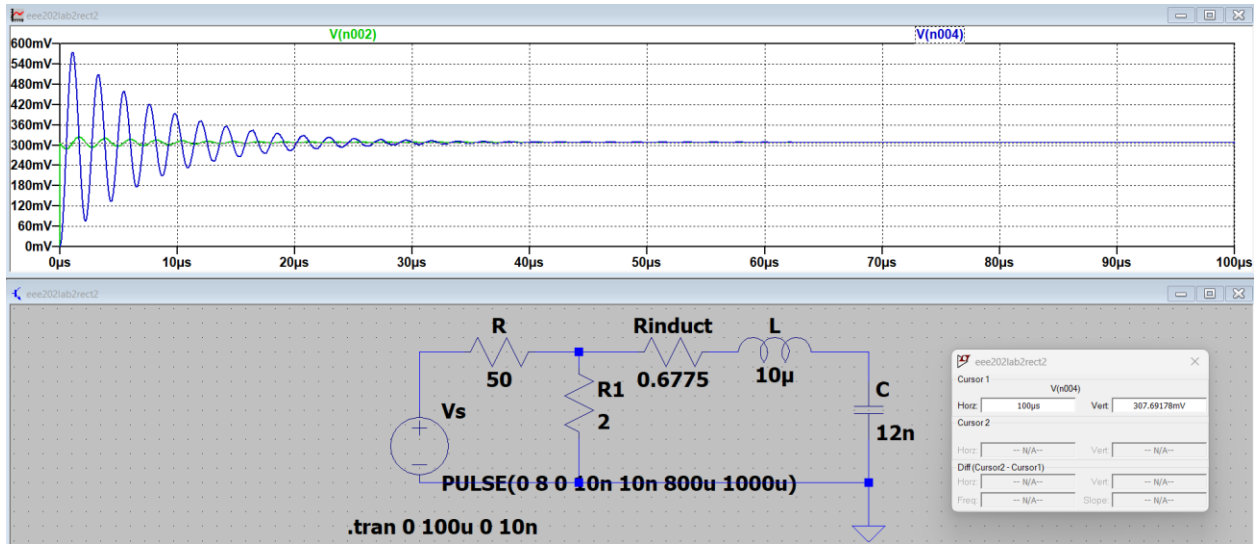


Figure C: LTSpice Simulation of our rectangular step circuit (Green = Input Voltage, Blue = Output Voltage)

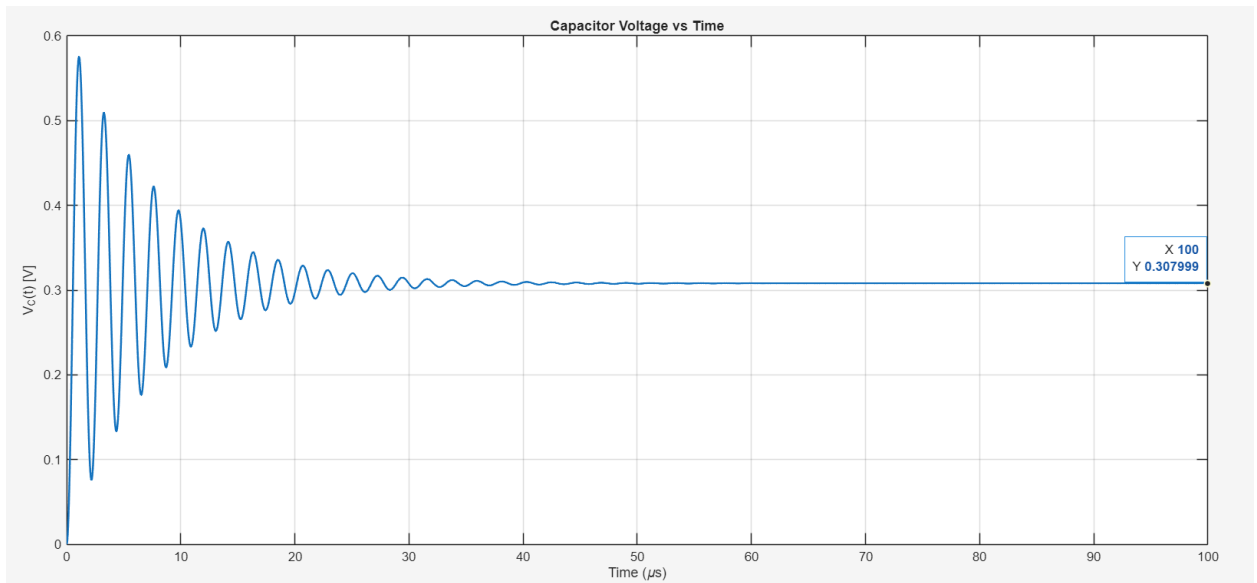


Figure D: MATLAB plot of our rectangular step circuit's analytical solution (Code in Appendix)

By inspection, it was observed that both graphs matched each other extremely closely. For an error calculation the endpoints of the waveform for our timesteps were picked. By comparing said points, the error was calculated to be **0.0997%** ($\frac{0.3076918 - 0.307999}{0.307999} \times 100$). Due to the shapes of both waveforms matching and the incredibly small error, which could be caused by floating-point rounding errors, this part was deemed to be successful.

In addition to the reaction to the steady state solution for an 8 Volt unit step forcing function, we needed to derive the behavior of the system for when the power supply voltage was set to zero at $t = T_1$. For this specific case, since T_1 has to be larger than QT_r where Q is the quality factor and T_r is the reciprocal of f_r , we found that T_1 has to be bigger than 24.1378 microseconds. So, T_1 was selected to be 40 microseconds. Since the system itself remained the same, the equation itself will remain the same as the solution for the homogenous case, as zero volts as a forcing function means that the solution is the general solution. Which means that the solution is in the form of equation (9). The only parameters that change are the initial conditions, which are voltage and current. Since we assume a steady state for both cases, we can say that the time-varying part of (12) equates to zero and the only part that remains is the constant voltage. Likewise, since the capacitor acts like an open circuit for a DC steady state, we can say that the current is also equal to zero. Therefore, we can say that the initial conditions are 0.308 Volts and 0 Amperes for $V_c(T_1)$ and $i_c(T_1)$ respectively. So, for the solution of this case, we only need to find A and B in equation (9). By applying the same process we did for equations (10) to (12), we can find A and B to be 0.308 and 0.0139 respectively and offsetting t by T_1 . Inserting these values into equation (9), we can plot the full function in MATLAB (Figure D.1) and compare it to LTSpice (Figure C.1).

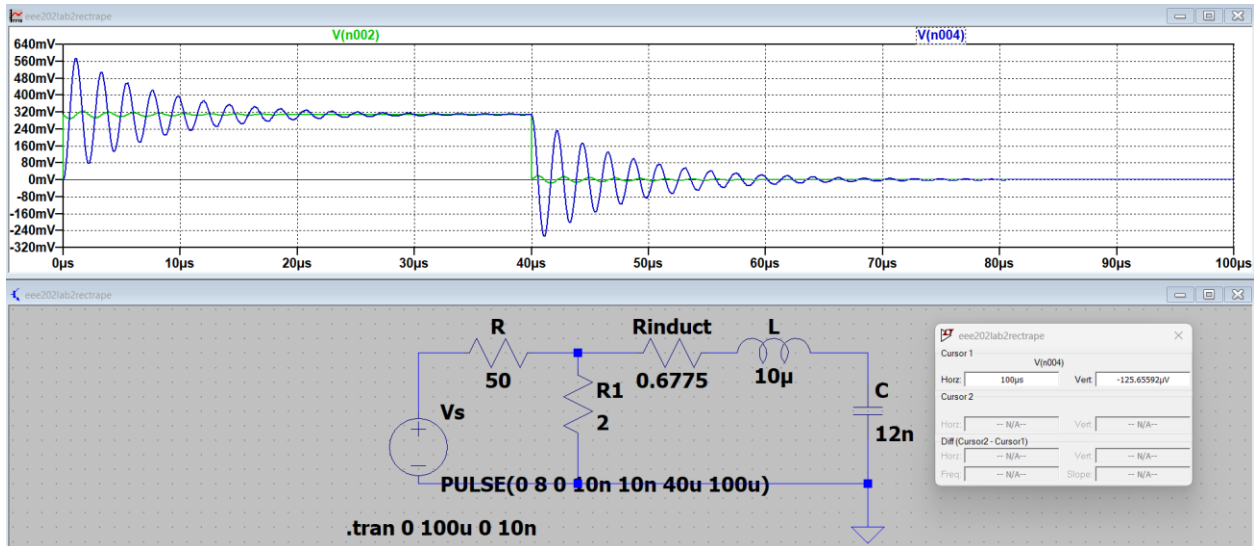


Figure C.1: LTSpice Simulation of our rectangular step circuit's zero result (Green = Input Voltage, Blue = Output Voltage)

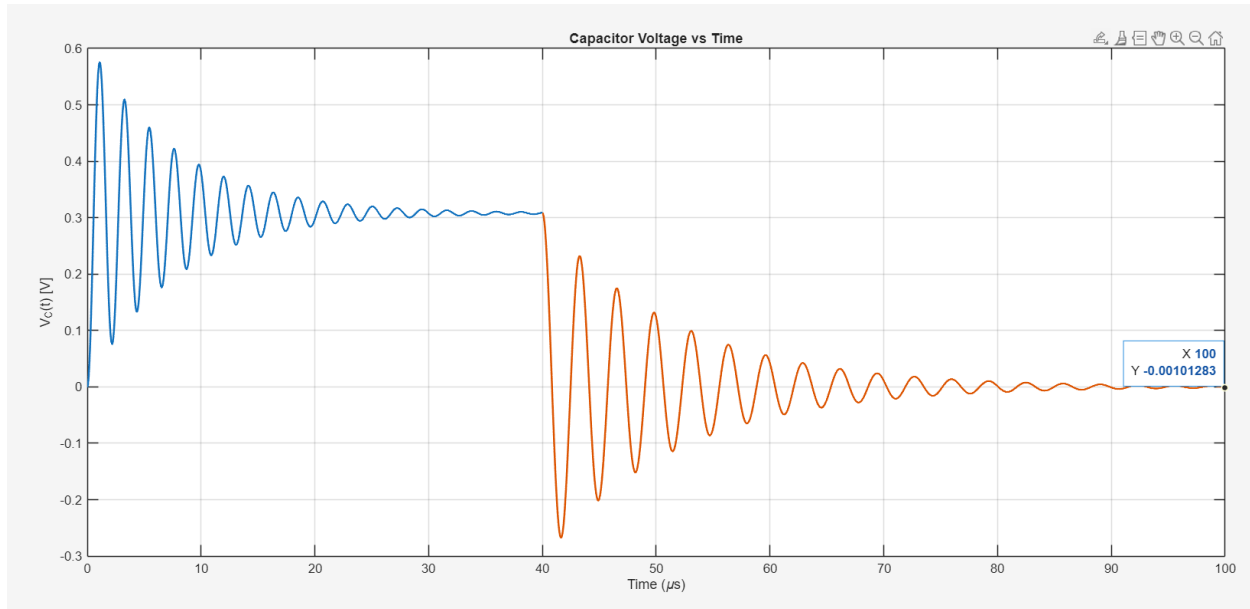


Figure D: MATLAB plot of our rectangular step circuit's zero result's analytical solution (Code in Appendix)

By inspection, it was observed that both graphs matched each other extremely closely. For an error calculation the endpoints of the waveform for our timesteps were picked. By comparing said points, the values found in both plots were incredibly close to zero, which is the expected form of a steady state waveform. Due to the shapes of both waveforms matching and the incredibly small error, which could be caused volatile behavior when functions approach zero, this part was deemed to be successful.

Preliminary Part 3 (Sinusoidal Wave Steady State Response):

In this part, we were asked to estimate the sinusoidal steady state response for our circuit in Figure A. Since the forcing function for this specific case can be written as $10\sin(2\pi f_r t)V$ since we will consider the steady state response for $t > 0$. Additionally, we will select an n greater than Q which will be the number of sinusoidal pulses. Since Q was found to be 11.0897, n was selected to be 20. For our analysis, due to the frequency being the resonant frequency, we can use Figure B and let the reactive components (the capacitor and the inductor) cancel each other out since in a series RLC circuit, we let the components cancel each other out. However, we need the capacitor voltage, but we can find it by analyzing the cancelled-out circuit, finding the current and integrating it whilst dividing it by the capacitance, as per the element parameter signifies that will give us the capacitor voltage.

For the cancelled-out circuit, we only have the source and the resistive components. In this case, for Figure B, V_x will be equal to $0.385\sin(2\pi f_r t)V$ since defined V_x to be $0.0385 V_s$ and since resistive components do not change the phase of the signal. Through this fact, we can find that the net current is approximately equal to $148.0428 \sin(2\pi f_r t)$ milliamperes as our net resistance is 2.6006 Ohms. By integrating and dividing in relation to the capacitor's element parameters, we can find capacitor voltage by the following derivation:

$$V_c(t) = \int \frac{i_c}{C} dt = \int \frac{148.0428 \sin(2\pi f_r t) \text{mA}}{12 \text{ nF}} dt = -4.2736 \cos(2\pi f_r t) V \quad (13)$$

Which is equal to the steady state response of the capacitor voltage for t in seconds. After this analytical solution was found, the circuit was formed in LTSpice and a transient analysis at a large enough

timeframe ($t > QT_r$) was performed to find the steady state voltage. The simulation was run and the input/output voltage was plotted (Figure E). Additionally, the analytical solution was written in MATLAB and the plot was observed (Figure F).

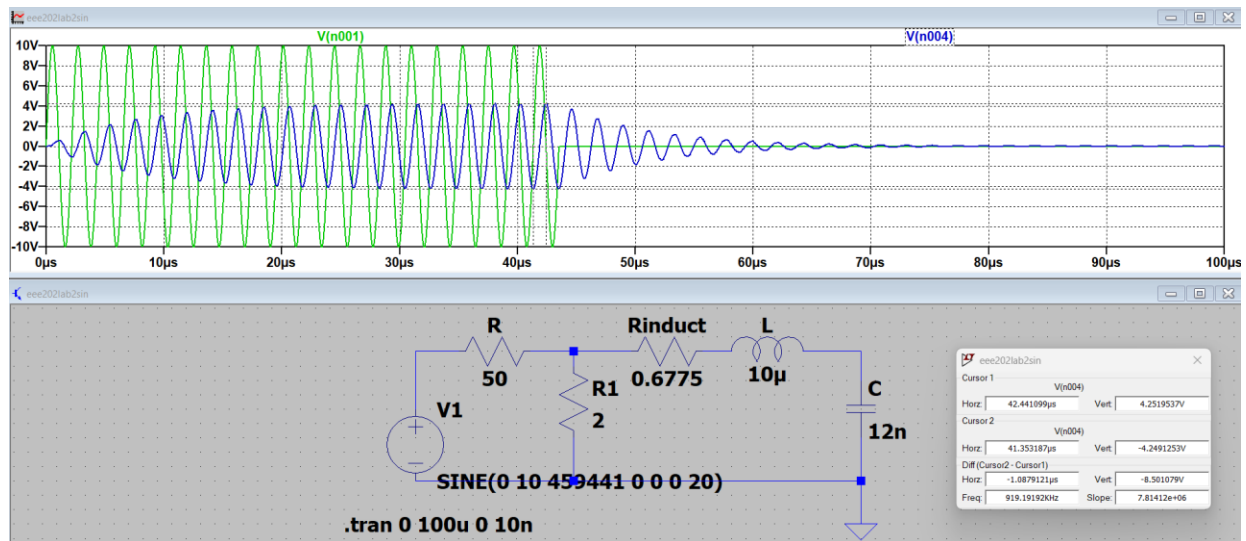


Figure E: LTSpice Simulation of our sinusoidal steady state (Green = Input Voltage, Blue = Output Voltage)

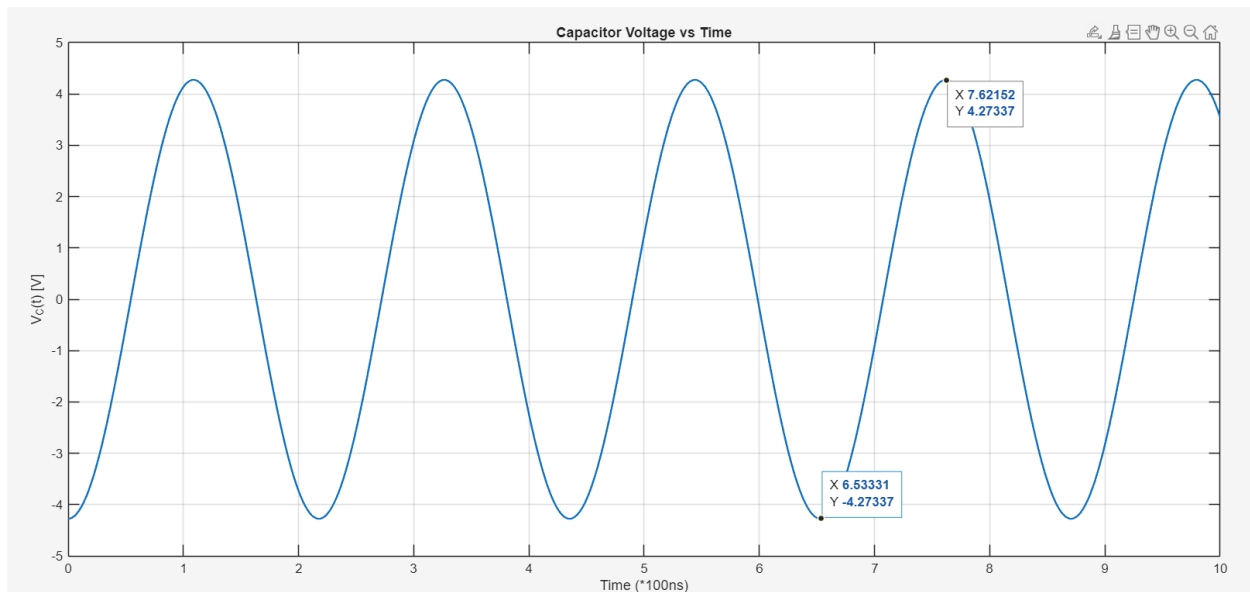


Figure F: MATLAB plot of our steady sinusoid circuit's analytical solution (Code in Appendix)

By inspection, it was observed that both graphs matched each other extremely closely. For an error calculation the maximum and minimum peaks of the waveform for our timesteps were picked. By comparing said points, the error was calculated to be **0.5342%** ($\frac{(4.27337+4.27337)-(4.25195+4.24913)}{4.27337+4.27337} \times 100$). Due to the shapes of both waveforms matching and the incredibly small error, which could be caused by floating-point rounding errors, this part was also deemed to be successful.

Preliminary Part 4 (Sinusoidal Transient Response):

For this part, we are asked to evaluate the transient response of the same circuit in part 3 for the time between zero and QT_r in LTSpice. Before inspecting LTSpice and its simulation, I have derived the analytical solution for the output voltage for positive time and put it in the Appendix.

For the simulation period, we need to know QT_r . Since we calculated Q in part 1 to be 11.0897 and T_r to be 2.1766 microseconds in the beginning of our analysis, we can determine QT_r to be approximately equal to 24.1378 microseconds. After calculating this value, the transient analysis simulation period was set to 24.1378 microseconds and the same circuit in part 3 was simulated. After the plot was observable, the input voltage, the capacitor voltage and the inductor current were plotted (Figure G).

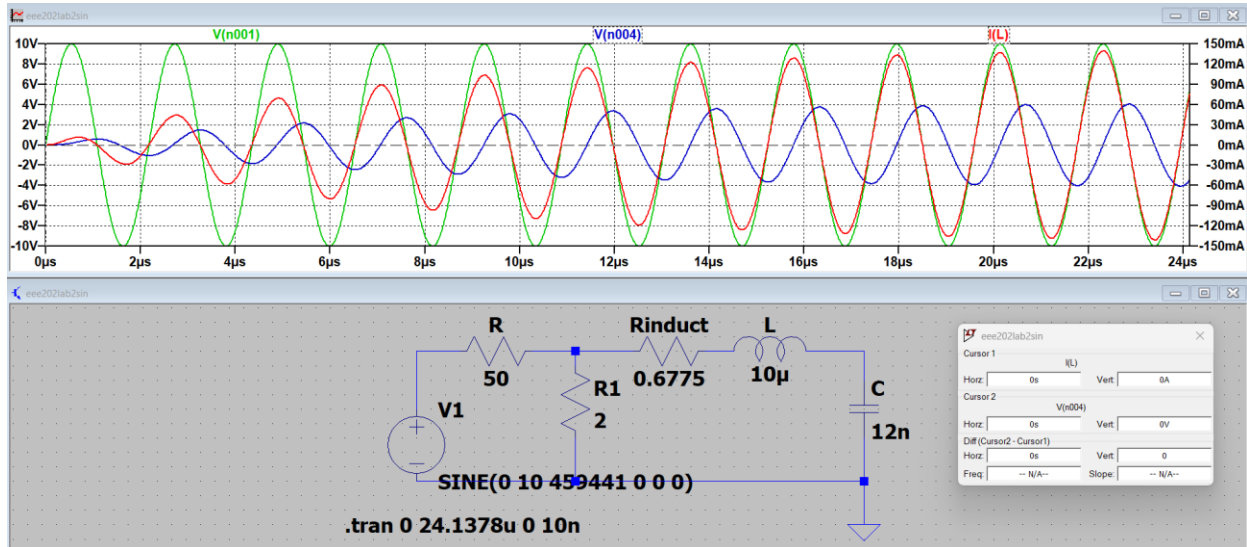


Figure G: LTSpice Simulation of our sinusoidal transient circuit (Green = Input Voltage, Blue = Output Voltage, Red = Current)

By inspection, it was found that the capacitor voltage was -90 degrees out of phase with the input voltage and that the inductor current had no phase-shift when compared to the input voltage. The reason for both of these can be the fact that we are at the resonant frequency, so that the current has no phase-shift. Since the current is a negative exponentially increasing sinusoid, and the capacitor voltage is related with the current's integral, a -90-degree phase-shift can be observed. Finally, the cursors at $t=0$ ensure that the initial conditions of zero capacitor voltage and zero inductor current have been met. Due to these facts, the simulation was deemed to be a success.

Hardware Implementation:

After the simulation and analytical results were gathered, the design was to be implemented. Before that, the time values T_0 and T_1 had to be determined. According to the Lab 2 document, T_1 has to be greater than QT_r where T_r is the resonant period – equal to 2.1766 microseconds as per the analysis section of this report – and also greater than $15T_r$. Due to this fact, T_1 was picked as 40 microseconds. Additionally, T_0 has to be bigger than $2T_1$, so T_0 was picked to be 100 microseconds. Finally, an n greater than the calculated quality factor had to be determined. Since our quality factor was approximated to be 11.0897, an n of 20 was picked.

After these values were determined, the design was implemented by soldering one 2.2 Ohm resistor whose value was extremely close to 2 Ohms (measured to be 2.05 Ohms), which was then soldered in

parallel with a branch of a series LC, where L was equal to 10 microhenries and C was equal to 12 nanoFarads (Figure 1.1). Through these steps, the hardware creation part was done.

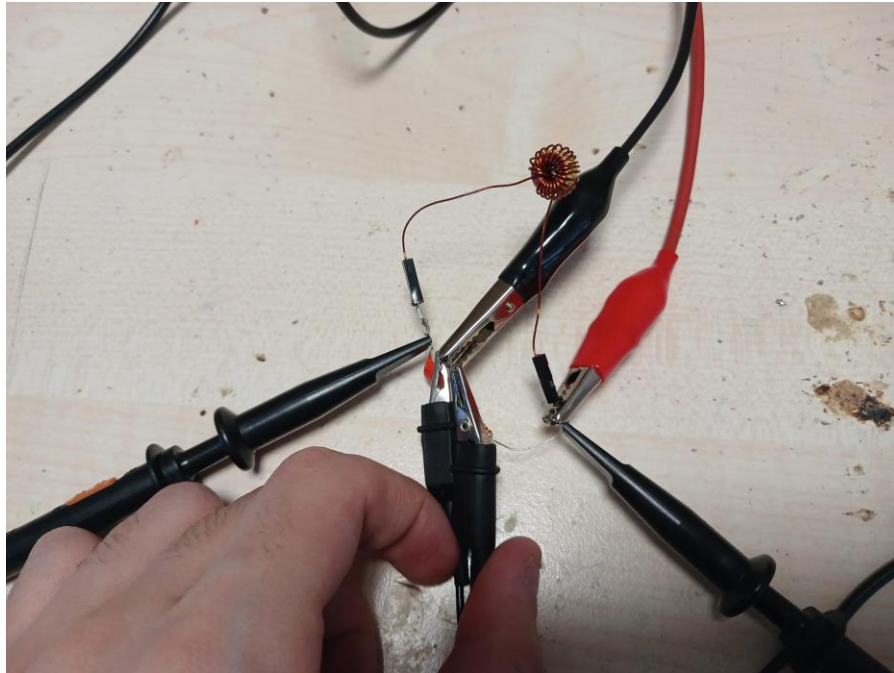


Figure 1.1: Implemented design with assigned component values with power connected

Using the formed design, power and probing was required. In order to match the schematic, the node between the inductor and resistor was supplied with power from the signal generator and the node between the capacitor and the resistor was grounded. For the waveform displays; one probe was connected to the power wire to view input voltage, and the other was connected to the node between the inductor and the capacitor to measure the output voltage, which was the capacitor voltage for this specific assignment. Both of the oscilloscope probes' ground clips were connected to the node between the resistor and the capacitor to have consistent grounding across the circuit. After these steps, the system was ready to receive signals to analyze.

Before sending signals to our implemented design, the oscilloscope and the signal generator had to be ready. The probe viewing the input voltage was connected to channel two, and the probe viewing the output voltage was connected to channel one. A coaxial cable was used to connect the TRIG OUTPUT terminal of the signal generator to the EXT TRIG terminal of the oscilloscope. The acquisition settings for the oscilloscope were set in accordance to the settings provided in the Lab 2 document. After finalizing the oscilloscope steps, the signal generator (whilst not connected to the circuit) was set with the parameters described in the Lab 2 document for the step input excitation phase to generate a periodic 8 Volt pulse supplier with pulse duration of T_1 and pulse period of T_0 . After all of these steps were followed, the signal generator's output terminal was connected to the circuit and the waveforms were recorded and analyzed (Figure 1.2, 1.3).

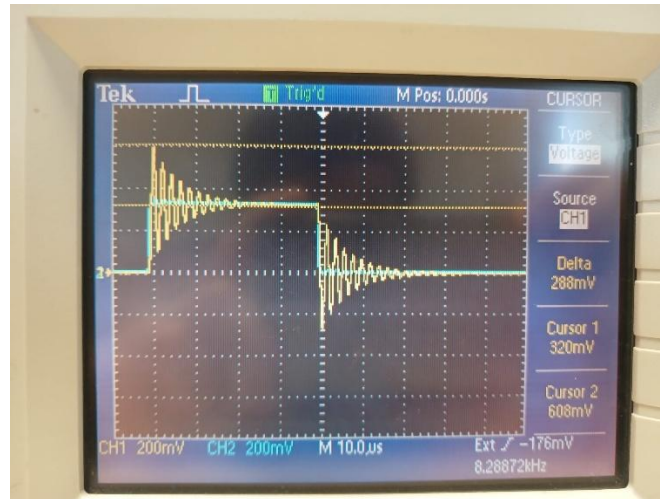


Figure 1.2: Implemented circuit receiving the defined rectangular step input voltage with steady state voltage and initial peak cursored (Yellow = Capacitor Voltage, Blue = Input Voltage)

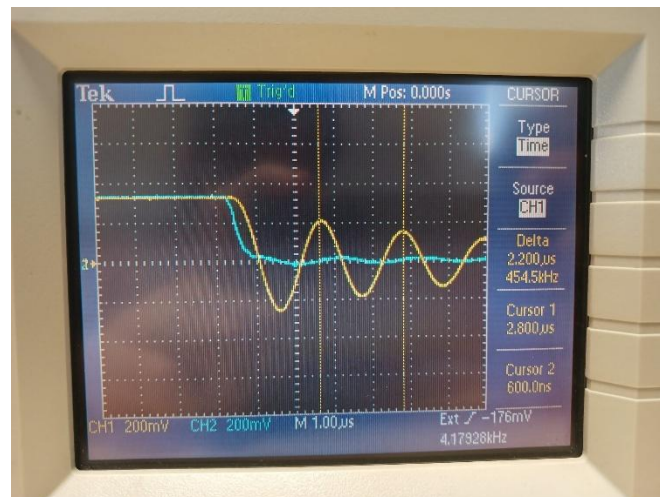


Figure 1.3: Implemented circuit receiving the defined rectangular step input voltage with peak-to-peak duration cursored (Yellow = Capacitor Voltage, Blue = Input Voltage)

First, the steady state voltage was measured to be 320 mV and the initial peak voltage was measured to be 608 mV. From the LTSpice simulations, these values were expected to be 307.69 mV and 574.69 mV respectively. Secondly, the sinusoidal enveloped signal's frequency was observed to be 454.5 KHz, which is extremely close to the resonant frequency of the circuit. All of these data points were recorded for future error calculations.

After the rectangular pulse input waveforms were observed, the power supply was configured in accordance to the sinusoidal pulse excitation section of the Lab 2 document. The implemented design and it's probing points were kept the same and the power was sent to the circuit. The gathered input and output voltage waveforms were recorded and analyzed (Figure 1.4, 1.5).

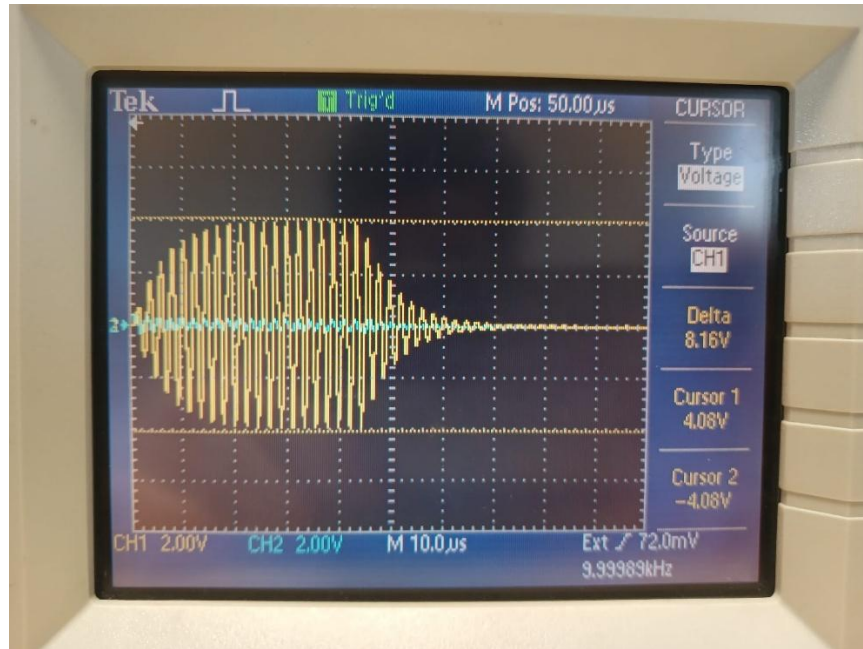


Figure 1.4: Implemented circuit receiving the defined sinusoidal pulse voltage with steady state voltage cursored (Yellow = Capacitor Voltage, Blue = Input Voltage)

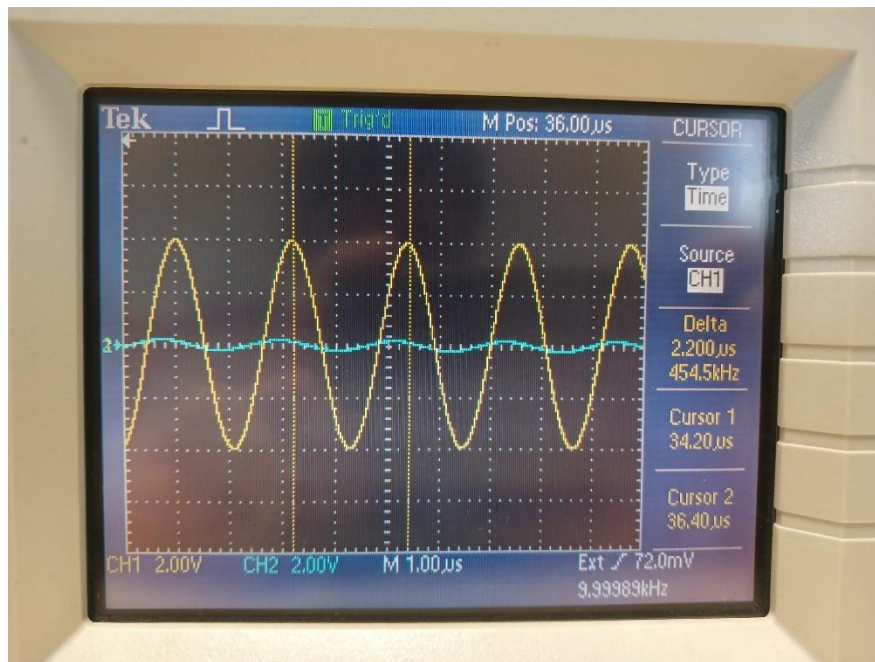


Figure 1.5: Implemented circuit receiving the defined sinusoidal pulse voltage with peak-to-peak duration cursored (Yellow = Capacitor Voltage, Blue = Input Voltage)

The steady state voltage was measured to be 8.16 Vpp. From the LTSpice simulations, this value was expected to be 8.50 Vpp. Additionally, the sinusoidal enveloped signal's frequency was observed to be 454.5 KHz, which is extremely close to the resonant frequency of the circuit. All of these data points were recorded for future error calculations.

Using the data recorded in the hardware part of the lab and the values derived in the preliminary work, a table was formed to compare each data point and their errors (Table 1). Errors were calculated by comparing simulative and real-life results. Each data entry was looked at with their errors. Some small error values were measured for the hardware implementation when compared to the analytical section. These could be the result of human error during measurement and tolerances that come with each component. Due to the small error values, the experiment was deemed to be a success.

	Output Voltage (V)	Time of Measurement (μ s)	Error (%)
MATLAB (Rectangular)	0.575226	1.0802	-
LTSpice (Rectangular)	0.5747322	1.0918	-
Real-Life (Rectangular)	0.608	1.080	5.4717
MATLAB (Rectangular)	0.308967	39.988	-
LTSpice (Rectangular)	0.3086656	39.9973	-
Real-Life (Rectangular)	0.320	39.80	3.5420
MATLAB (Sinusoidal)	8.5471 (Peak to Peak)	42.70 – 43.60	-
LTSpice (Sinusoidal)	8.5062 (Peak to Peak)	42.44 - 43.53	-
Real-Life (Sinusoidal)	8.16 (Peak to Peak)	34.20 – 35.30	4.0700

Table 1: Collection of analytical, simulative and real-life values and errors

Conclusion:

The purpose of this lab assignment was to understand the behavior of a second-order circuit with resistive, inductive and capacitive components. First, the general solution to the transient part of the output signals were derived analytically and the specific cases were calculated with specific initial values and assigned component values. Then, the quality factor of our circuit was determined and found to indicate that the behavior of the circuit is underdamped, which means that we will have oscillatory behavior before reaching a steady state. After that, the capacitor voltage's transient behavior for an unit step voltage input of magnitude 8 Volts, steady state behavior for a sinusoidal wave of 10 Volts for positive time and the transient behavior of the same sinusoidal input was derived analytically. The analytical derivations were plotted using MATLAB and the circuits with said initial conditions were simulated using LTSpice. For both preliminary cases that required both LTSpice and MATLAB, the plots matched incredibly closely. There was a small calculated error for both cases, but due to the small nature of the errors, it can be said that they are due to floating-point errors due to the precision of our plots. From these results, we have gathered very strong comparisons for the real-life experiment for the unit step and sinusoidal pulses for the second-order circuit respectively.

After the simulative and analytical parts, the design was created in real life to get actual measurements. When the experiment was carried out, certain values were observed. These measurements were recorded and compared to the theoretical values to find the error. The error was calculated and was found to be extremely small relative to the measurements. Through this lab assignment, RLC Voltage response, circuit

analysis derivations and hardware implementation skills were gained and improved; especially for sinusoidal and rectangular pulses.

References:

R. E. Thomas, A. J. Rosa, and G. J. Toussaint, *The Analysis and Design of Linear Circuits*. Hoboken, NJ, USA: Wiley.

H. Koymen and A. Atalar, *Analog Electronics*, ver. 4.2. Ankara, Turkey: Bilkent University, 2015.

Appendix:

Code for Step Input Solution (Figure D):

```
clear
clc
close all

t = linspace(0, 100e-6, 5000);

Vc = -exp(-1.3003e5 .* t) .* ...
      (0.308*cos(2.8838e6 .* t) + 0.0139*sin(2.8838e6 .* t)) ...
      + 0.308;

figure;
plot(t*1e6, Vc, 'LineWidth', 1.5);
xlabel('Time (\mus)');
ylabel('V_C(t) [V]');
title('Capacitor Voltage vs Time');
grid on;
```

Code for Step Input Zero Solution (Figure D.1):

```
clear
clc
close all
```

```

t = linspace(0, 40e-6, 5000);
tnext = linspace(40e-6, 100e-6, 5000)

Vc = -exp(-1.3003e5 .* t) .* ...
    (0.308*cos(2.8838e6 .* t) + 0.0139*sin(2.8838e6 .* t)) ...
    + 0.308;
Vcnext = exp(-1.3003e5 .* t) .* ...
    (0.308*cos(2.8838e6 .* t) + 0.0139*sin(2.8838e6 .* t))

figure;
plot(t*1e6, Vc, 'LineWidth', 1.5);
hold on;
plot((tnext*1e6), Vcnext, 'LineWidth', 1.5);
xlabel('Time (\mus)');
ylabel('V_C(t) [V]');
title('Capacitor Voltage vs Time');
grid on;

```

Code for Sinusoidal Steady State Solution (Figure F):

```

clear
clc
close all

t = linspace(0, 100e-7, 5000);

L= 10e-6;
C= 12e-9;
w= 1/sqrt(L*C);

Vc = -4.2736*cos(w*t);

```

```

figure;
plot(t*1e6, Vc, 'LineWidth', 1.5);
xlabel('Time (*100ns)');
ylabel('V_C(t) [V]');
title('Capacitor Voltage vs Time');
grid on;

```

Analytical Solution for Preliminary Part 4:

First of all, since we computed Q and T_r to be equal to 11.0897 and 2.1766 microseconds in the beginning of our analysis, we can determine QT_r to be equal to approximately 24.1378 microseconds. Additionally, since we need to find a solution for this case, we need to find the homogenous case -which we already did- and the particular solution. Since we are working with positive time, we can reuse the same forcing function that we used in part 3 ($0.385\sin(2\pi f_r t)V$) as our source voltage for the Figure B circuit. To find the particular solution, we would normally need to write the particular capacitor voltage in terms of $V_{cp}(t) = A\sin(2\pi f_r t) + B\cos(2\pi f_r t)V$ and solve equation (4) with this to find A and B. However, since the homogenous case exponentially goes to zero by inspecting equation (9), we can say that the homogenous part does not affect our steady state; and since we found the steady state response in part 3 as equation (13), which must be the particular solution. Therefore, we can determine the solution for the sinusoid forced function case to be:

$$V_C(t) = e^{-1.3003 \times 10^5 t} [A \cos(2.8838 \times 10^6 t) + B \sin(2.8838 \times 10^6 t)] V - 4.2736 \cos(2\pi f_r t) V \quad (14)$$

Which is valid for all time greater than zero in seconds.

Now, we need to find the coefficients A and B in terms of our initial conditions. As is tasked in the Lab 2 document, we assume, at $t=0$, zero capacitor voltage and zero inductor (i.e. net) current. By inputting zero into equation (14), we see that A is equal to the forced function's amplitude, which is 4.2736 Volts. By utilizing the same trick we did in part 2 (taking the derivative, setting it to zero at $t=0$ and solving it since initial current is said to be zero), we can take the derivative of equation (14) and write it as:

$$\begin{aligned} \frac{dV_C(t)}{dt} = & (e^{-1.3003 \times 10^5 t} \times 2.8838 \times 10^6) [-A \sin(2.8838 \times 10^6 t) + B \cos(2.8838 \times 10^6 t)] - \\ & (1.3003 \times 10^6 \times e^{-1.3003 \times 10^5 t}) [A \cos(2.8838 \times 10^6 t) + B \sin(2.8838 \times 10^6 t)] V + \\ & 4.2736 \sin(2\pi f_r t) V \quad (15) \end{aligned}$$

By utilizing this equation and our determined value of A, we can find that B is equal to -0.193 Volts. Now that we know both A and B for our conditions, the solution for our specific case can be written as:

$$\begin{aligned} V_C(t) = & e^{-1.3003 \times 10^5 t} [4.2736 \cos(2.8838 \times 10^6 t) + 0.193 \sin(2.8838 \times 10^6 t)] V \\ & - 4.2736 \cos(2\pi f_r t) V \quad (12) \end{aligned}$$

For $t > 0$ in seconds.